INCH-POUND

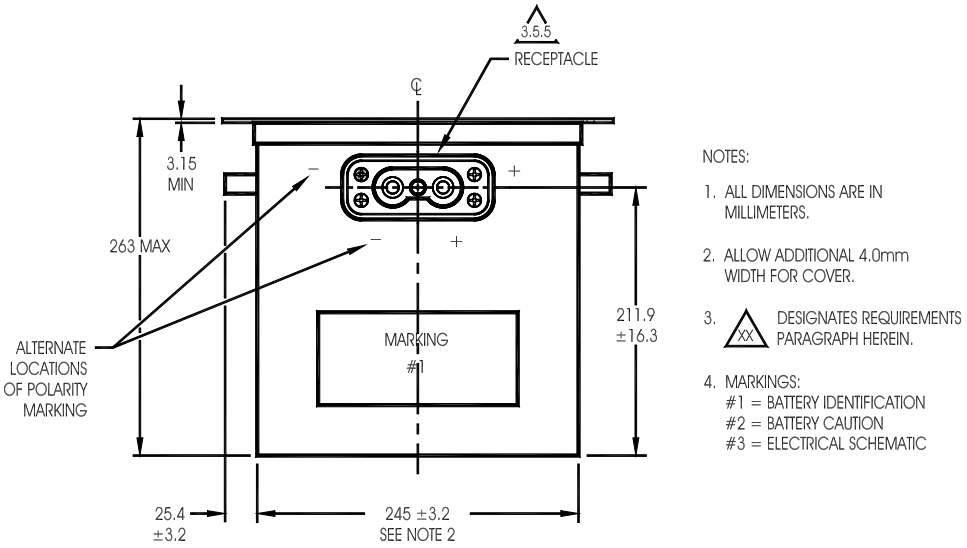
MIL-PRF-8565/5B
w/AMENDMENT 2
05 January 2022
SUPERSEDING
MIL-PRF-8565/5B(AS)
w/AMENDMENT 1
27 September 2010

PERFORMANCE SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, MAINTENANCE-FREE, 24-VOLT, 30-AMPERE-HOUR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-8565.



FRONT VIEW

FIGURE 1. Dimensions and configuration for PIN D8565/5-1.

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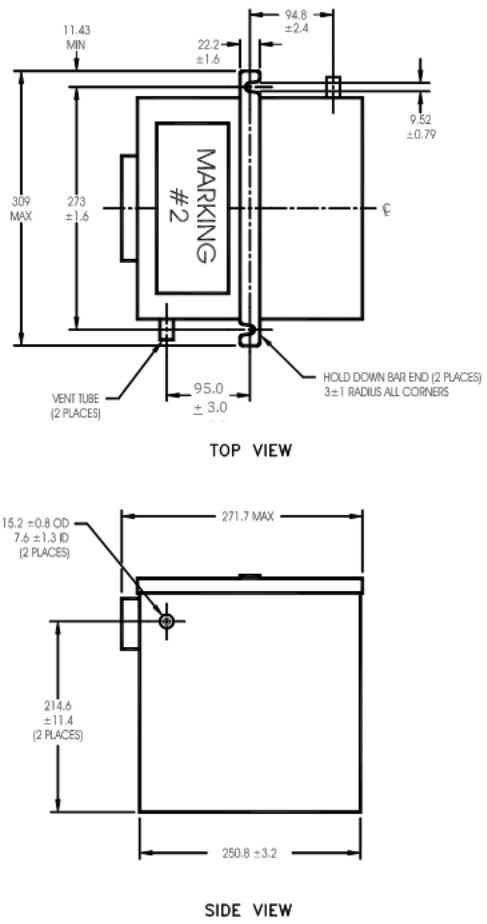
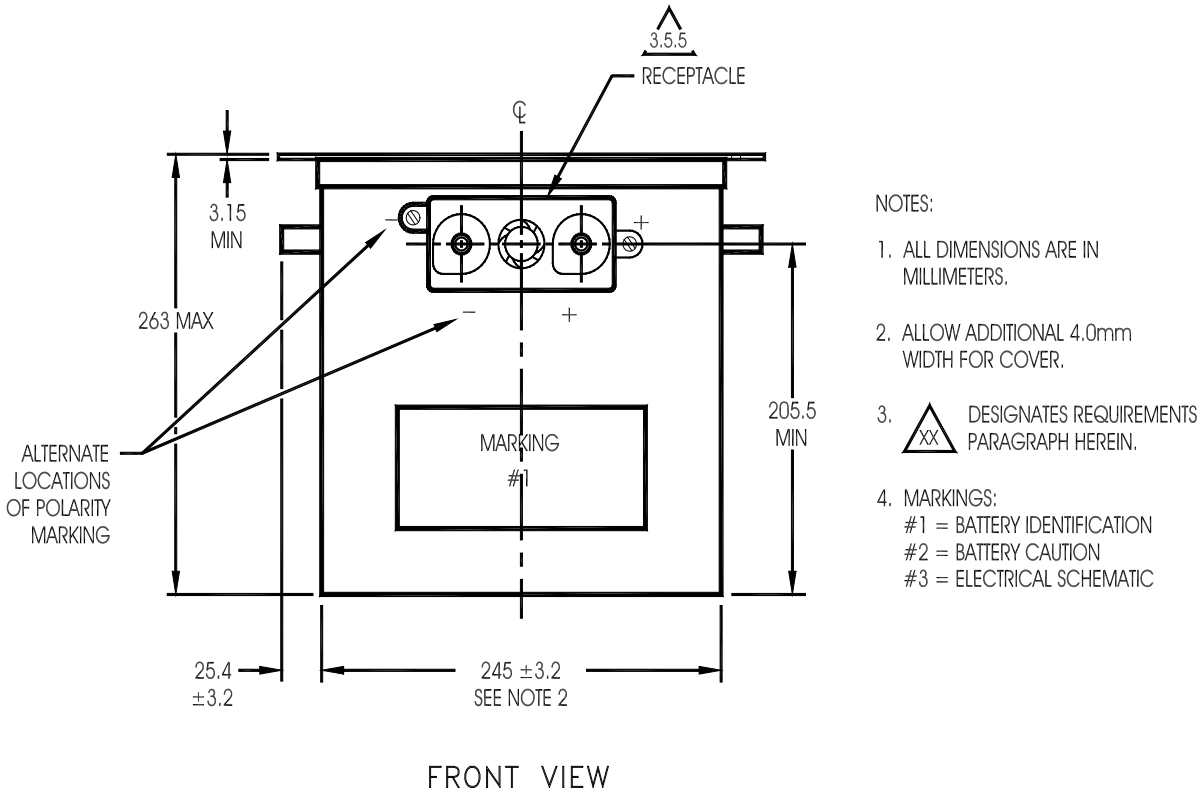


FIGURE 1. Dimensions and configuration for PIN D8565/5-1 - Continued.

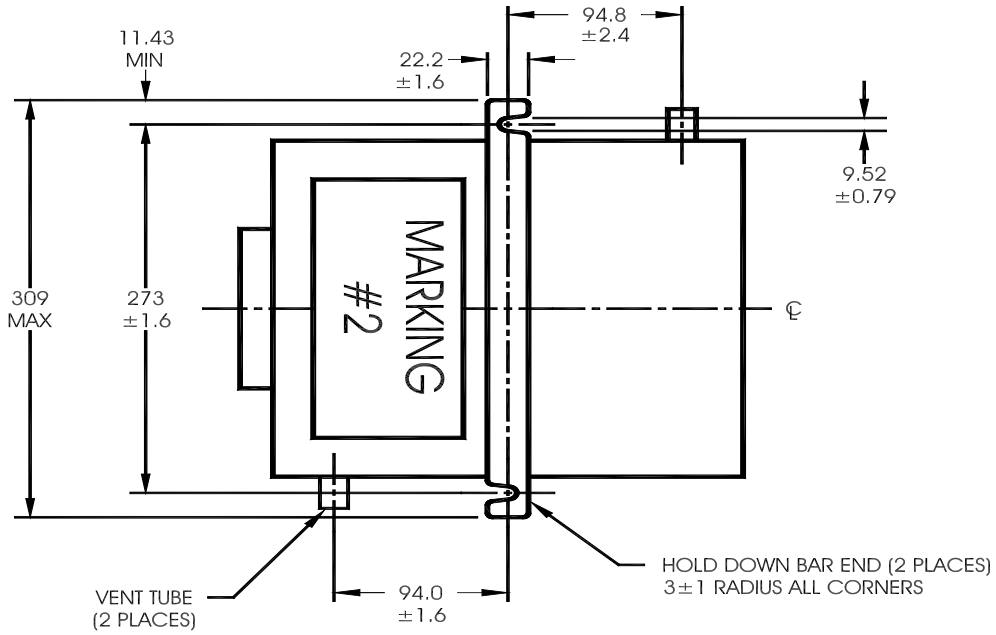
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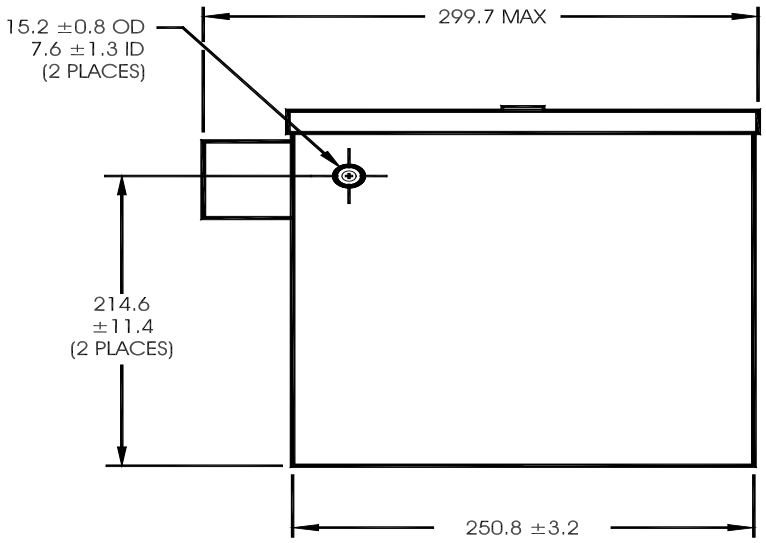
FRONT VIEW

FIGURE 2. Dimensions and configuration for PIN D8565/5-2.

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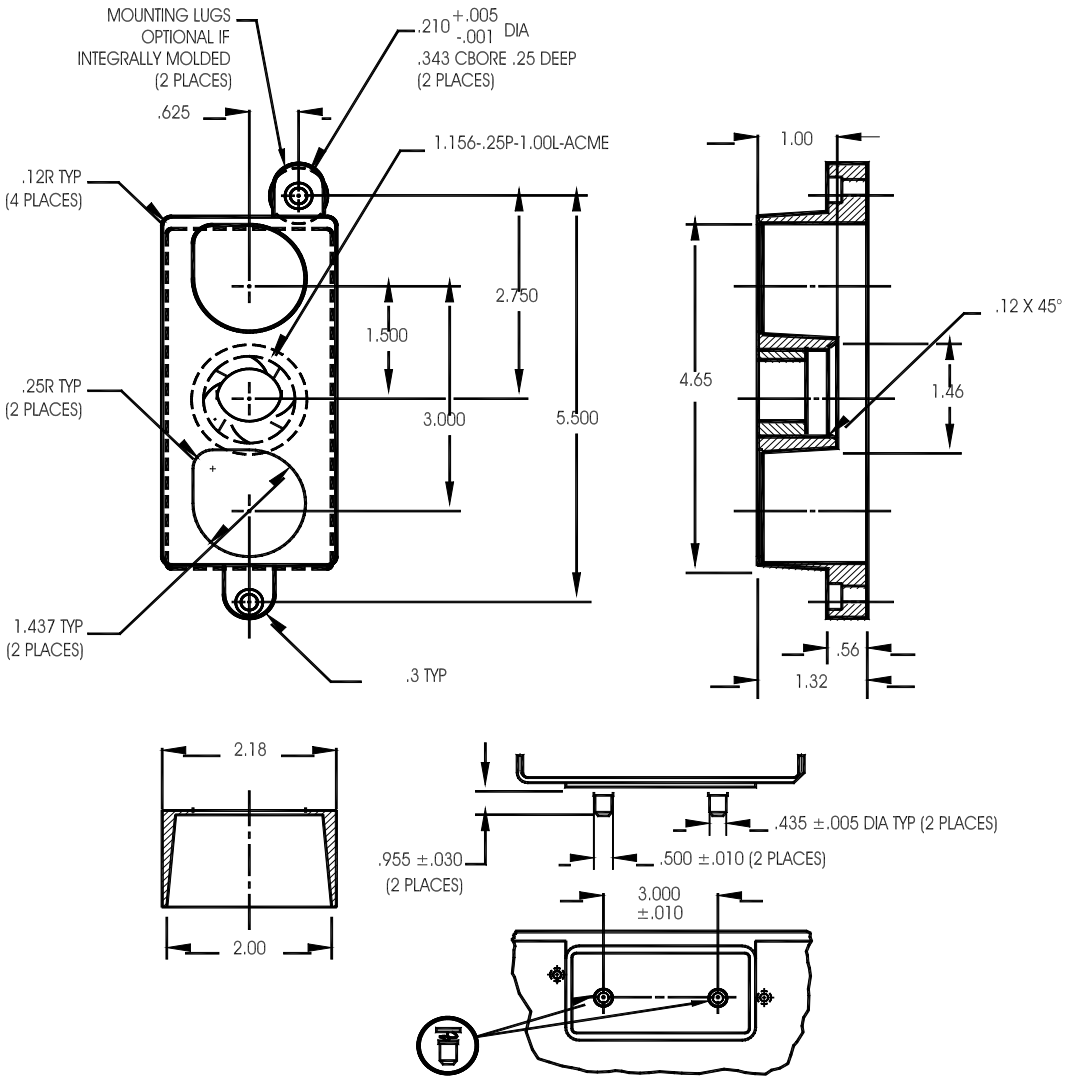
TOP VIEW



SIDE VIEW

FIGURE 2. Dimensions and configuration for PIN D8565/5-2 - Continued.

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NOTES:

1. MATERIAL OR FINISH TO BE ACID-RESISTANT.
2. DIMENSIONS ARE IN INCHES.
3. DIMENSION TOLERANCES ARE:
   X.XX ±0.03
   X.XXX ±0.010
4. RECEPTACLE TO CASE ATTACHMENT METHOD OPTIONAL FOR NONMETALLIC CONTAINERS.
5. DRAFT ANGLES ALLOWED IN CONFORMANCE WITH COMMERCIAL DIE CASTING MACHINES.

FIGURE 3. Typical receptacle adapter.

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REQUIREMENTS:

1. Part or identifying number. The part or identifying number (PIN) of the batteries covered by this specification sheet shall be D8565/5-1 for figure 1 and D8565/5-2 for figure 2.

2. Dimensions and weight. The dimensions of the battery shall be as shown on figures 1 and 2 of the specification sheet. The weight of the battery shall be not greater than 36.4 kilograms (see 3.8.3 and 4.5.3). In addition, the weight of any production battery shall vary by not greater than ± 1.0 kg from the average weight of the three approved qualification sample batteries but may not exceed the 36.4-kilogram limit.

3. Capacity. The rated capacity of the battery shall be not less than 30.0 ampere-hours (Ah). See 3.8.6 as modified herein.

4. Service life. The battery's service life (useful life) in aircraft shall be not less than 3 years. The battery shall require no scheduled maintenance during the 3 years (see 3.8.9, 4.5.10, and 6.15.10).

5. Qualification inspection and conformance inspection. Table I lists the required inspections.

6. MIL-PRF-8565 variance. The battery shall comply with MIL-PRF-8565 except as follows.

6.1 The following paragraphs are not applicable:
- 3.6.5 Marking for batteries with heaters.
- 3.8.16 Output performance.
- 3.8.17 Cold and hot temperature charge.
- 4.5.28 Output performance test.
- 4.5.29 Cold and hot temperature charge test.

6.2 Modify the following paragraphs:

3.5.5 Receptacles. Insert the following after the first sentence: "For D8565/5-1, the female receptacle (front side) shall be dimensionally equal to MIL-PRF-18148/3. For D8565/5-2, the female receptacle shall be dimensionally equal to figure 3 of the specification sheet. The receptacle may be integrally molded in the container."

3.8.6 Capacity and electrical performance. Delete the text and insert: "The rated capacity of the battery shall be not less than 30.0 ampere-hours. To meet the capacity requirement, each battery shall produce a potential of not less than 18.0 volts at not less than 60 minutes when connected to a 30.0-ampere load after being tested in accordance with 4.5.7 at 21 °C (70 °F) (30.0 Ah/1.0 hr/21°C/18.0 V at end of life). The battery's 1C-rate shall be 30 amperes (see requirement 3 of the specification sheet)."

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3.8.7 Constant voltage discharge. Delete the first sentence and insert: "The battery shall produce not less than the current specified below after being tested in accordance with 4.5.8."

Temperature	Current (amperes) at 0.3 seconds	Current (amperes) at 15 seconds	Current (amperes) at 30 seconds	Average current over the total 60 seconds
Room temperature (see 4.4.1)	875	750	700	684
-26.0 ± 2.0 °C	625	500	420	430

4.5.10 Cycling test. In the third sentence, change "2.0-hour" to "4.0-hour".

6.3 Add the following paragraph:

"6.20 PIN supersession data. PIN supersession information and cross-references are indicated below:

"D8565/5-1 supersedes M83769/5-1 and M83769/5-2.
"D8565/5-2 supersedes M83769/1-1 and M83769/6-2."

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TABLE I. Qualification inspection and conformance inspection of batteries.

Test number	Examinations and tests	Qualification inspection					Conformance inspection		
		Sample number			Requirement paragraph	Test paragraph	Group A sample 1 100%	Group B sample number	
		1	2	3				1	2
1	INCOMING INSPECTION	X	X	X	3.8.1	4.5.1	X	X	X
2	VISUAL AND MECHANICAL EXAMINATION	X	X	X	3.5, 3.6, 3.8.2, table II	4.5.2	X	X	X
3	DIMENSIONS AND WEIGHT	X	X	X	Reqmt 2, 3.8.3	4.5.3	X 1/	X	
4	DIELECTRIC STRENGTH AND INSULATION RESISTANCE TEST	X	X	X	3.7, 3.8.4	4.5.4	X 1/	X	X
5	STRENGTH OF VENT TUBES TEST	X	X	X	3.7, 3.8.5	4.5.5	X 1/	X	X
6	ELECTRICAL TEST PREPARATION	X	X	X	---	4.5.6		X	X
7	CAPACITY DISCHARGE	X	X	X	Reqmt 3, 3.8.6	4.5.7	X	X	X
8	CONSTANT VOLTAGE DISCHARGE	X	X		3.7, 3,8,7, reqmt (4) of table III	4.5.8	X	X	
9	STRENGTH OF RECEPTACLE TEST			X	3.7, 3.8.8	4.5.9			X
10	CYCLING TEST	X			3.8.9, reqmt (1) of table III	4.5.10		X	
11	CAPACITY DISCHARGE AT 49°C (120°F)		X		3.8.6, reqmt (2) of table III	4.5.11			X
12	CHARGE AND DISCHARGE TEST AT LOW TEMPERATURE		X		3.8.6, reqmt (3) of table III	4.5.12			X
13	DISCHARGE WHILE INVERTED		X		3.5.9, reqmt (5) of table III	4.5.13			X
14	ALTITUDE TEST		X		3.7, 3.5.9, 3.8.10	4.5.14			X
15	SHOCK TEST (BASIC DESIGN)	X			3.7, 3.8.10	4.5.15		X	

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TABLE I. Qualification inspection and conformance inspection of batteries – Continued.

Test number	Examinations and tests	Qualification inspection					Conformance inspection		
		Sample number			Requirement paragraph	Test paragraph	Group A sample 1 100%	Group B sample number	
		1	2	3				1	2
16	TEMPERATURE SHOCK TEST	X			3.7, 3.8.10	4.5.16			
17	TEMPERATURE RISE AND FLOAT TEST		X		3.7, 3.8.11, reqmt (1) of table III	4.5.17			X
18	BATTERY GAS EMISSION TEST		X		3.8.10, 3.18.12	4.5.18			
19	VIBRATION TEST		X		3.7, 3.8.10	4.5.19			
20	HUMIDITY TEST		X		3.7, 3.8.10	4.5.20			
21	SALT FOG TEST	X			3.7, 3.8.10	4.5.21			
22	GROUND STORAGE TEST			X 2/	3.8.13.1	4.5.22			
23	DEEP DISCHARGE RECOVERY TEST			X	3.7, 3.8.13.3	4.5.27			
24	PHYSICAL INTEGRITY AT HIGH TEMPERATURE TEST	X			3.7, 3.8.10, reqmt (1) of table III	4.5.23			
25	SHOCK TEST (CRASH SAFETY)			X	3.8.10, 3.8.15	4.5.25			
26	IMPACT RESISTANCE TEST (NONMETALLIC BATTERY CONTAINER)		3/		3.8.14	4.5.24			
27	WORMGEAR RETAINING PIN HARNESS (D8565/1 ONLY)		4/		3.5.5	4.5.30			
28	SHELF LIFE TEST			X	3.8.9, 3.8.13.2	4.5.26			
29	FINAL EXAMINATION	X	X	X	3.7, 3.8.4	4.5.31		X	X

1/ See sample information in test paragraph.
2/ The manufacturer requesting qualification to this specification may submit an additional (fourth) test sample to expedite the qualification test program.
3/ For nonmetallic battery container designs, two outer nonmetallic battery containers shall be furnished for inspection.
4/ Five wormgear retaining pins shall be furnished for inspection.

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Referenced documents: In addition to MIL-PRF-8565, this document references MIL-PRF-18148/3.

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

CONCLUDING MATERIAL

Custodians:
Navy - AS
Air Force - 170
DLA - CC

Preparing activity:
Navy - AS

(Project 6140-2021-020)

Review activity:
Air Force - 70

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